Full support and positive density in the zero-temperature SK model

One lower bound, two consequences · A reconstruction of OpenAI's argument · 24 October 2026

Abstract. The Sherrington–Kirkpatrick (SK) model assigns a random energy to each list of signs, called a spin configuration. The Parisi formula expresses its limiting maximum energy per spin as a minimum over nondecreasing functions. For this pair-interaction model with no external field, every integrable minimiser has an increment measure with full support on $[0, 1)$: every interval of positive length receives positive mass. This measure also has a smooth density, strictly positive on $(0, 1)$. We reconstruct OpenAI's argument for these two conclusions. A gap in the measure's support would be an interval where the minimising function is constant. The proof associates a curve with the minimiser. Minimality forces this curve to equal time at the gap's endpoints. A quantitative lower bound makes its derivative strictly convex on a positive constant interval, producing an impossible area deficit; zero coefficients and endpoints have separate arguments. Once all gaps are excluded, the same bound passes through an approximation at a fixed time and forces the density to be positive. Chen had already proved full support and smooth absolute continuity. The contribution here is an account of the two uses of the quantitative bound, with their analytic inputs and attribution retained.

1. The result and the mechanism

A configuration of n spins is a list of signs $\sigma_1, \dots, \sigma_n \in \{-1, 1\}$. For each pair $i < j$, choose an independent Gaussian weight J_{ij} with mean zero and variance one; these weights are called couplings. The Sherrington–Kirkpatrick optimization problem is to maximise the energy

$$H_n(\sigma) = n^{-1/2} \sum_{i<j} J_{ij}\sigma_i\sigma_j, \qquad \sigma \in \{-1, 1\}^n.$$

Write $P_* := \lim_{n\to\infty} n^{-1}\mathbb{E} \max_\sigma H_n(\sigma)$ for the limiting optimum per spin. The Parisi formula expresses this value as a minimum over nondecreasing functions (see Auffinger and Chen (2017), Theorem 1 and proof of Lemma 3). The parameter t ranges over nonnegative overlap scales: for two spin configurations σ, τ, their overlap $R = n^{-1} \sum_i \sigma_i\tau_i$ measures their agreement, with $R = 1$ when they coincide and $R = -1$ when every sign is reversed. We explain why a minimising function has no interval of constancy, and why the measure defined by its increments has positive density in the interior.

This is the analytic part of one proof about the quantum approximate optimization algorithm (QAOA), which prepares spin configurations by a sequence of quantum operations. For each desired accuracy, that proof chooses one finite sequence whose expected energy per spin approaches P_* to that accuracy as $n \to \infty$. The absence of intervals on which the minimising function is constant supplies identities for expectations along a Brownian-driven random process. These identities construct a near-optimal classical input for the quantum operations. Positive density is a further conclusion. Section 5 explains this connection and points to the separate circuit argument. The proofs here use Itô calculus and standard martingale convergence, with the required estimates for the Parisi partial differential equation (PDE) stated where they enter.

The model has only pair interactions (pure SK) and no one-spin bias term (zero external field). Its covariance function is $\xi(t) = t^2/2$: for configurations with overlap R, $\mathbb{E}[H_n(\sigma)H_n(\tau)] = nR^2/2-1/2$. Adding the same independent centred Gaussian of variance $1/2$ to every configuration's energy gives covariance $n\xi(R)$ and leaves the expected maximum unchanged. This matches the convention in the Parisi formula.

Let $\mathcal{U}$ consist of the nonnegative, nondecreasing, right-continuous functions $\gamma : [0,1) \to [0,\infty)$ satisfying $\int_0^1 \gamma(t)\,dt < \infty$. Define the locally finite Stieltjes measure by

$$\mu_\gamma([0,t]) = \gamma(t), \qquad 0 \le t < 1.$$

Thus $\mu_\gamma((s,t]) = \gamma(t) - \gamma(s)$: the measure records the increments of γ. A point mass (atom) of size $\gamma(0)$ is allowed at zero, and the total mass need not be finite or equal to one. The support consists of points whose every neighbourhood in $[0,1)$ has positive mass. A support gap is a relatively open interval of positive length with zero mass, on which γ is constant. When the measure has a density ρ, its mass on an interval is the integral of ρ there. All support statements are relative to $[0,1)$.

The Parisi functional $\mathcal{P}(\gamma)$ is defined using an auxiliary value function $\Phi_\gamma(t,x)$, where $x \in \mathbb{R}$ is a spatial variable and t is the overlap parameter, treated as time in the equation. Write $\Phi = \Phi_\gamma$ and use subscripts for partial derivatives. The backward equation, terminal condition and functional are

$$\Phi_t = -\tfrac{1}{2}(\Phi_{xx} + \gamma(t)\Phi_x^2), \qquad \Phi(1,x) = |x|, \qquad \mathcal{P}(\gamma) = \Phi_\gamma(0,0) - \tfrac{1}{2}\int_0^1 t\gamma(t)\,dt.$$

For an integrable coefficient, the solution is interpreted by approximation from bounded monotone coefficients. The exact control definition and stability theorem are OpenAI (2026a), §1 and Lemmas 2.1–2.2. To turn minimality of $\mathcal{P}$ into a condition on γ, we use a random process whose gradient moment measures the first-order cost of changing γ. Write $u = \Phi_x$ for the spatial gradient and $a = \Phi_{xx}$ for its spatial derivative. Let B be standard Brownian motion and define the diffusion X and the moment q by

$$dX_t = \gamma(t)u(t,X_t)\,dt + dB_t, \qquad X_0 = 0, \qquad q(t) = \mathbb{E}[u(t,X_t)^2].$$

Thus q is the expected square of the gradient along the diffusion. The diffusion exists and is unique for the fixed Brownian motion. We use stochastic calculus on compact time intervals below one; the required analytic inputs are stated in §2.

Chen (2026), Theorem 1.3 had already proved full support and smooth absolute continuity. The following formulation and quantitative proof are from OpenAI; the positive-density consequence is treated separately in §4.

Theorem 1 (full support; OpenAI (2026a), Theorem 1.1). If $\gamma \in \mathcal{U}$ minimises $\mathcal{P}$, then

$$\operatorname{supp}\mu_\gamma = [0,1), \qquad \gamma(t_2) > \gamma(t_1) \quad (0 \le t_1 < t_2 < 1).$$

For every associated diffusion and every $0 \le t < 1$,

$$q(t) = t, \qquad \mathbb{E}[a(t,X_t)^2] = 1.$$

Its Stieltjes measure has a smooth density, strictly positive at every interior time; Corollary 3 gives the precise stronger statement. Existence of a minimiser is a separate input from Auffinger and Chen (2017), Theorem 1 and the proof of Lemma 3. Integrability allows γ to grow near one. Zero field gives $u(0,0) = 0$, and $\xi'' = 1$ is used in the curvature calculation; the theorem here does not extend these hypotheses to nonzero field or a general mixed-spin covariance.

The proof compares two demands on a hypothetical gap. Minimality forces q to agree with time at its endpoints, with endpoint slopes at most one. On a positive constant interval a quantitative

estimate makes q' strictly convex, so its integral is smaller than the interval's length. This contradicts the required change in q. After excluding all gaps, the identity $q' = 1$ supplies the equation from which the same estimate will force $\gamma' > 0$.

The support conclusion has a history distinct from this proof. Auffinger, Chen and Zeng (2020), Theorem 1 proved infinite support, which can still have gaps: consider $\sum_{k\geq 1} 2^{-k}\delta_{1-2^{-k}}$, where δ_x is unit mass at the point x. We follow OpenAI's quantitative argument and credit its antecedents at the steps where they enter.

2. Minimality gives contact; curvature excludes gaps

For an admissible coefficient, Φ is even, convex and one-Lipschitz in space. Its positive-order spatial derivatives are bounded and continuous on each strip $[0,T]\times\mathbb{R}$, $T<1$; the PDE holds almost everywhere in time. The compact-strip approximation theorem justifies the following Itô calculations even when γ has jumps (see OpenAI (2026a), Lemmas 2.1–2.2). The control method has antecedents in Jagannath and Tobasco (2016) at positive temperature and Chen, Handschy and Lerman (2018), Proposition 2 and Theorem 5 at zero temperature.

Fix a minimiser. Itô's formula cancels the drift of $u(t,X_t)$:

$$du(t,X_t) = a(t,X_t)\,dB_t, \qquad q(0)=0, \qquad q'(t) = \mathbb{E}[a(t,X_t)^2].$$

In particular q' is continuous across coefficient jumps. Minimality means that every allowed one-sided change has nonnegative first-order cost. The one-sided variation formula gives the rate of change of the functional when γ is replaced by $\gamma+\varepsilon\eta$ as $\varepsilon\downarrow 0$:

$$D_+\mathcal{P}(\gamma)[\eta] = \tfrac{1}{2}\int_0^1 \eta(t)(q(t)-t)\,dt, \tag{1}$$

for every integrable direction η for which $\gamma+\varepsilon\eta\in\mathcal{U}$ for all sufficiently small $\varepsilon\geq 0$ (see OpenAI (2026a), Lemma 6.1). Earlier zero-temperature variation and support identities appear in Chen, Handschy and Lerman (2018), Propositions 3–4.

Adding an atom at s changes the cumulative function by $\eta = \mathbf{1}_{[s,1)}$. Thus

$$G(s) := \int_s^1 (q(t)-t)\,dt \geq 0.$$

Here $G(s)/2$ is the first-order cost of the added atom. Conversely, let ν be the restriction of μ_γ to a compact $K\subset[0,1)$. Removing a fraction of this mass is feasible, so

$$0 \leq D_+\mathcal{P}(\gamma)[-\nu([0,\cdot])] = -\tfrac{1}{2}\int_K G(s)\,d\mu_\gamma(s).$$

The change of integration order is justified by $|q-t|\leq 1$ and $\mu_\gamma(K)<\infty$. Since $G\geq 0$, it vanishes almost everywhere for μ_γ, and continuity makes it vanish on $S=\operatorname{supp}\mu_\gamma$. This removes diffuse mass as well as atoms; no density or probability constraint is assumed.

At a support point $s>0$, G has a minimum. Hence

$$G'(s) = s-q(s) = 0, \qquad G''(s) = 1-q'(s)\geq 0, \quad \text{so} \quad q(s)=s, \quad q'(s)\leq 1. \tag{2}$$

If $0 \in S$, then $G(0) = q(0) = 0$ and $G(h) = \frac{1}{2}(1 - q'(0))h^2 + o(h^2)$; nonnegativity gives the same slope bound at zero. These are the contact conditions of OpenAI (2026a), Lemma 6.2.

Lemma 2 (constant-interval curvature; OpenAI (2026a), Proposition 5.1). Let $\gamma \in \mathcal{U}$, with its associated diffusion and moment q. If $\gamma = c$ on an open interval $I \subset (0,1)$, called a plateau, then for $t \in I$

$$q'''(t) \geq \frac{c^2}{1000}\mathbb{E}[a(t, X_t)^4] > 0 \quad \text{if } c > 0. \tag{3}$$

If $c = 0$, then $q''(t) = \mathbb{E}[\Phi_{xxx}(t, X_t)^2] > 0$.

Minimality is not a hypothesis of this lemma. Its integrated coercive estimate is explained in §3. We first show what it buys.

Suppose (ℓ, r) is a support gap with endpoints in S and $r < 1$. The coefficient is constant there. If its value is positive, strict convexity of q' and (2) give

$$q'(t) < \frac{r - t}{r - \ell}q'(\ell) + \frac{t - \ell}{r - \ell}q'(r) \leq 1 \qquad (\ell < t < r).$$

Integrating contradicts $q(r) - q(\ell) = r - \ell$. The geometric obstruction is an area deficit under q'. If the constant is zero, strict increase gives $q'(t) < q'(r) \leq 1$ and the same contradiction. An initial gap $[0, r)$ with $0 \notin S$, $r \in S$ and $r < 1$ has coefficient zero and no atom at zero. Use $q(0) = 0$ and contact at r, without assuming contact at zero.

A terminal gap cannot use contact at one, which is outside the support domain. On a constant interval ending at one, OpenAI (2026a), Lemma 6.3 instead proves

$$1 - q(t) \geq C\sqrt{1 - t}$$

for some $C > 0$ and t sufficiently close to one. Evolving the terminal condition $|x|$ backwards smooths its corner at zero. On the assumed plateau, this keeps its gradient u uniformly away from ± 1 in a window of width $\sqrt{1-t}$ around zero. The diffusion density has a uniform positive lower bound on that window, under the assumed plateau; integrating $1 - u^2$ gives the estimate. Thus $q(t) - t \leq (1-t) - C\sqrt{1-t} < 0$ near one, making $G(s) < 0$ there. This contradicts minimality and also excludes empty support. The zero-coefficient heat argument and late-step variation already appear in Chen, Handschy and Lerman (2018), Theorem 6 and its proof. Terminal-plateau exclusion was already recorded by Auffinger, Chen and Zeng (2020), Remark 7.

These cases exhaust the complement of the relatively closed set S. Consequently $S = [0, 1)$, contact gives $q(t) = t$, and differentiation gives $\mathbb{E}a^2 = 1$, including the right-hand value at zero. Moreover, $\gamma(t_2) - \gamma(t_1) = \mu_\gamma((t_1, t_2]) > 0$. This proves Theorem 1 using the stated analytic inputs.

3. The estimate behind strict curvature

On a plateau with $\gamma = c > 0$, put $r = c\Phi_x$, $v = r_x$, $w = r_{xx}$ and $z = r_{xxx}$. In expectations below, these functions are evaluated at (t, X_t). Along the diffusion, the drift and diffusion pairs for r, v, w are $(0, v)$, $(-v^2, w)$ and $(-3vw, z)$. Starting with $c^2 q' = \mathbb{E}v^2$, Itô's formula gives

$$c^2 q'' = \mathbb{E}[w^2 - 2v^3], \qquad c^2 q''' = \mathbb{E}[z^2 - 12vw^2 + 6v^4]. \tag{4}$$

For example, the drifts of w^2 and v^3 are $z^2 - 6vw^2$ and $3vw^2 - 3v^4$, respectively. The negative mixed term is the obstacle. Convexity, which gives $v \geq 0$, does not control it.

The required estimate is an inequality after integration against the actual diffusion density. We first prove it for finite-step coefficients: on each constant step the equation can be transformed into a heat equation, making the needed shape restrictions accessible. We then pass to general integrable coefficients by approximation. Begin with a positive nondecreasing finite-step coefficient γ_{step}: a function constant on each interval of a finite partition. Replace the terminal value $|x|$ by the smooth approximation $M^{-1}\log(2\cosh Mx)$, with $M \geq \|\gamma_{\mathrm{step}}\|_\infty$. At a positive time on a step of height c, let f be the probability density of X_t. It factors as $f = e^{c\Phi}g$, where g solves the forward heat equation. For $s = -\partial_x \log g$, the source proves joint shape restrictions: on $x > 0$, r and s are increasing and concave, $w/(2r) + v \leq 0$ is nondecreasing, and s/r is nondecreasing. The ratios extend through zero. These are analytic conclusions of OpenAI (2026a), §3, not consequences of the polynomial in (4).

The factorisation of the density gives $f_x/f = r - s$. For a suitable expression F in the shape quantities, define

$$\mathcal{W}(F) = r\partial_x F + (v + r^2 - rs)F.$$

Then $f\mathcal{W}(F) = \partial_x(rfF)$, so the source's tail estimates give $\mathbb{E}_f\mathcal{W}(F) = 0$, where $\mathbb{E}_f$ denotes integration against $f(x)\,dx$. The point is to choose F so that subtracting this zero-mean correction from $z^2 - 12vw^2 + 6v^4$ permits a decomposition into a square and terms nonnegative under the joint shape restrictions, with remainder at least $10^{-3}v^4$. The correction changes the integrand while preserving its expectation. The conclusion is

$$\int_{\mathbb{R}} (z^2 - 12vw^2 + 6v^4) f\,dx \geq \frac{1}{1000}\int_{\mathbb{R}} v^4 f\,dx. \tag{5}$$

The exact identity is OpenAI (2026a), Lemma 4.1; the shape hypotheses, correction and boundary justification are assembled in its Proposition 4.2. The long reasoning record, §4.4, "How the signs make a correction with zero expectation useful", defines every auxiliary variable and reproduces the certificate. Thus (5) is a cited analytic input here, with its mechanism exposed; it is not a pointwise assertion for arbitrary v, w, z.

To obtain Lemma 2 for an integrable coefficient, choose positive monotone step approximations retaining the plateau near a fixed interior time, and send the terminal-smoothing parameter to infinity. Spatial derivatives through order four converge uniformly on a fixed strip below one, and the diffusions driven by the same Brownian motion converge (see OpenAI (2026a), Lemmas 2.1–2.2 and Proposition 5.1). Only v, w, z remain in (5), so no convergence theorem for the auxiliary logarithmic derivative s is needed. Dividing by c^2 gives (3); strictness follows from $\Phi_{xx} > 0$ on the plateau (see OpenAI (2026a), Lemma 2.3). The zero plateau has the separate heat identity in Lemma 2. Its strictness uses a positive diffusion density and the fact that Φ_{xxx} cannot vanish identically: an even quadratic profile would contradict the one-Lipschitz bound and $|x| + O(1)$ growth.

The exponential change of variables that converts each constant-step equation to a heat equation (the Cole–Hopf representation), together with the rules joining successive steps, has antecedents in Lopatto (2026), Propositions 3.6, 7.1 and 9.1 and Chen (2026), §§3–4. OpenAI's estimate also uses the joint restriction $r\,\partial_x(s/r) \geq 0$, beyond separate monotonicity and concavity. Chen's Proposition 4.2 proves the crossing implication $q''(t) = 0 \Rightarrow q'''(t) > 0$ on a hypothetical positive gap of the minimiser. In contrast, (3) is quantitative and holds throughout every positive plateau of any admissible coefficient. Chen's Appendix C distinguishes earlier shared estimates from the parallel crossing proofs.

4. A second use: the density cannot vanish inside the interval

Full support does not force pointwise positive density. For instance, $\gamma(t) = \int_0^t (s - \frac{1}{2})^2\, ds$ has full Stieltjes support but $\gamma'(\frac{1}{2}) = 0$. The quantitative remainder in (5) rules this out for a minimiser.

Corollary 3 (smooth positive density; OpenAI (2026a), Corollary 6.4). Under the hypotheses of Theorem 1, $\gamma(0) = 0$ and $\gamma \in C^\infty([0,1))$, with derivatives at zero taken from the right. Its Stieltjes measure is $\mu_\gamma(dt) = \gamma'(t)\, dt$, and

$$2\gamma'(t)\mathbb{E}[a(t, X_t)^3] \geq \frac{\gamma(t)^2}{1000}\mathbb{E}[a(t, X_t)^4] > 0 \qquad (0 < t < 1). \tag{6}$$

Only nonnegativity of the density at zero is asserted.

Write $b = \Phi_{xxx}$ and $d = \Phi_{xxxx}$; all spatial factors in expectations below are evaluated at (t, X_t). Since $da = -\gamma a^2\, dt + b\, dB$, Itô's formula applied to a^2 gives, for almost every t, $0 = \frac{d}{dt}\mathbb{E}a^2 = \mathbb{E}b^2 - 2\gamma\mathbb{E}a^3$. Here the left side vanishes because full support gives $\mathbb{E}a^2 = 1$. Solving for the coefficient yields

$$\gamma(t) = \frac{\mathbb{E}b^2}{2\mathbb{E}a^3}. \tag{7}$$

Initially (7) holds almost everywhere. Its denominator is at least two, because $a \geq 0$ and $\mathbb{E}a^3 \geq (\mathbb{E}a^2)^{3/2} = 1$. The moments are continuous; right continuity of γ extends (7) to every time and proves continuity of γ. Evenness gives $b(0,0) = 0$, hence $\gamma(0) = 0$.

For smoothness, consider expectations of polynomials in finitely many spatial derivatives of Φ; these are the polynomial moments. Apply Itô's formula to this family of moments at once. The derivative of each moment is another such moment plus γ times such a moment. Continuity makes these moments C^1; (7) then makes γ C^1. Repeating proves smoothness without presupposing time derivatives of the coefficient. This is the quotient-and-moment argument of Chen (2026), Proposition 4.6 and Appendix B.6, using the polynomial-moment method of Auffinger and Chen (2015), Lemma 2 and the proof of Theorem 2(ii).

Now the two differentiated moments are

$$\frac{d}{dt}\mathbb{E}b^2 = \mathbb{E}[d^2 - 6\gamma ab^2], \qquad \frac{d}{dt}\mathbb{E}a^3 = \mathbb{E}[3ab^2 - 3\gamma a^4].$$

Differentiating $\mathbb{E}b^2 = 2\gamma\mathbb{E}a^3$ gives

$$2\gamma'\mathbb{E}a^3 = \mathbb{E}[d^2 - 12\gamma ab^2 + 6\gamma^2 a^4]. \tag{8}$$

The expression from the plateau calculation has returned. The minimiser is not constant on an interval, so applying the plateau lemma directly would be invalid. Here is the needed transfer.

Fix $t \in (0,1)$ and $c = \gamma(t) > 0$. For $0 < \delta < \min(t, 1-t)$ define

$$\gamma_\delta(s) = \begin{cases} c, & t - \delta \leq s < t + \delta, \\ \gamma(s), & \text{otherwise.} \end{cases}$$

Both joins preserve monotonicity, and this remains an admissible coefficient. Moreover,

$$\|\gamma_\delta - \gamma\|_1 \leq 2\delta\,[\gamma(t+\delta) - \gamma(t-\delta)] \longrightarrow 0.$$

Use a subscript δ for its value-function derivatives and diffusion. Lemma 2 and (4), already extended to integrable coefficients, give at the fixed time t

$$\mathbb{E}[d_\delta^2 - 12c\, a_\delta b_\delta^2 + 6c^2 a_\delta^4] \geq \frac{c^2}{1000}\mathbb{E}[a_\delta^4]. \tag{9}$$

Choose a single T with $t < T < 1$. Compact-strip derivative stability and convergence of the diffusions driven by the same Brownian motion pass each spatial moment in (9) to its counterpart for γ; the estimates do not depend on the plateau width (see OpenAI (2026a), Lemmas 2.1–2.2 and Corollary 6.4). Combining the limiting inequality with (8) proves (6). Strictness follows from $c > 0$ and $\mathbb{E}a^4 \geq (\mathbb{E}a^2)^2 = 1$. In particular, Cauchy–Schwarz gives $\mathbb{E}a^3 \leq \sqrt{\mathbb{E}a^4} \leq \mathbb{E}a^4$, so the same inequality implies $\gamma'(t) \geq \gamma(t)^2/2000$ for $0 < t < 1$.

We have passed spatial moments, not the time derivatives q_δ'''. Indeed the limiting minimiser has $q(t) = t$ and $q''' = 0$; the coefficient derivative in (8) is exactly what the constant-interval formula omits. The long record, §§5.1–5.2, retains the full moment bootstrap and fixed-time transfer.

5. The scalar consequence and a separate circuit theorem

Full support also recovers the scalar value P_* defined in §1. This maximum-energy limit is called the ground-state value in the sign convention used here. The argument does not use the density corollary. For a minimiser, the ground-state Parisi formula gives $P_* = \mathcal{P}(\gamma)$ (see Auffinger and Chen (2017), Theorem 1).

The bounded martingale $M_t = u(t, X_t) = \int_0^t a\, dB$ has $\mathbb{E}M_t^2 = t$. It converges almost surely and in L^2 to a variable U. Since $|M_t| \leq 1$ and $\mathbb{E}M_t^2 \to 1$, we have $|U| \leq 1$ and $\mathbb{E}U^2 = 1$, hence $|U| = 1$ almost surely. The bound $|\gamma u| \leq \gamma$ gives $X_t \to X_1$ almost surely and in L^2. The terminal estimate

$$0 \leq \Phi_\gamma(t, x) - |x| \leq \sqrt{2(1-t)/\pi} + \tfrac{1}{2}\int_t^1 \gamma(s)\, ds$$

and the convex difference-quotient bounds force $u(t, X_t) \to \operatorname{sign}(X_1)$ on $\{X_1 \neq 0\}$. Thus $UX_1 = |X_1|$, also when $X_1 = 0$. Itô's formula for $\Phi(t, X_t)$ and passage to time one yield $\mathcal{P}(\gamma) = \mathbb{E}|X_1| - \int_0^1 t\gamma(t)\, dt$. Multiplying $X_1 = B_1 + \int_0^1 \gamma M_t\, dt$ by U then cancels the penalty, because $\mathbb{E}[UM_t] = t$:

$$P_* = \mathbb{E}[UB_1] = \int_0^1 \mathbb{E}[a(t, X_t)]\, dt. \tag{10}$$

Fubini uses $|UM_t| \leq 1$ and $\gamma \in L^1$; the final covariance identity for stochastic integrals uses $\mathbb{E}\int_0^1 a^2\, dt = 1$. This first-moment integral is different from $\mathbb{E}a^2 = 1$. The argument is OpenAI (2026a), Corollary 7.1; the long record, §6, "Recovering the value from a martingale", supplies the endpoint details. The gradient martingale and moment identities have antecedents in Montanari, preprint v2, Lemmas 3.2–3.4, under its positive-temperature support assumption. The zero-temperature martingale and value calculations are in El Alaoui, Montanari and Sellke, preprint v1, Lemmas 6.5 and 6.13. These are method antecedents, distinct from the finite-Gaussian construction below.

Here is the role of full support in the circuit proof. The identity $\mathbb{E}a^2 = 1$ permits normalised Gaussian increments weighted by $a = \Phi_{xx}$, while (10) identifies their target value. At fixed $T < 1$, discretising the diffusion gives a terminal gradient depending on finitely many Gaussians. This

gradient lies in $[-1, 1]$, whereas a spin must be -1 or 1. Rounding with an independent random variable produces such a sign-valued (Boolean) output while preserving the gradient as its conditional mean, and therefore preserving its correlations with the preceding Gaussian variables. A specified sum of products of its correlations with the driving Gaussians and the normalised increments converges to $\int_0^T \mathbb{E}a(t, X_t)\, dt$ as the mesh is refined; taking $T \uparrow 1$ then brings its value to P_*. This is the additional finite construction of OpenAI (2026a), Proposition 7.2, reconstructed in the long record, §7, "A finite Gaussian spin and its adjacent correlations".

There is also a separate circuit theorem. QAOA begins with equal quantum amplitudes on all spin configurations. Each layer alternates a phase change determined by H_n (cost evolution) and single-spin rotations that mix the signs (transverse-field mixing). The depth is the number of layers; their angles specify the strengths of these operations (see Farhi, Goldstone and Gutmann (2014)). For every $\varepsilon > 0$, OpenAI (2026b), Theorem 1.1 constructs a finite depth and deterministic angles, independent of system size n and the couplings J_{ij}, whose limiting expected energy per spin is at least $P_* - \varepsilon$. The angles and depth may depend on ε; the limit $n \to \infty$ is taken with them fixed. Fixed-parameter limits are due to Farhi, Goldstone, Gutmann and Zhou (2022), whose arbitrary-fixed-angle theorem uses this Hamiltonian and gate convention directly. The eventual-optimality conclusion was conjectured by Basso et al. (2022), §7, equation (7.2). The circuit argument must preserve that correlation value, not just reproduce a scalar number. It realises the finite Gaussian construction in a recursive space organised by a root and its descendants, approximates the bounded output by quantum gates, removes auxiliary random inputs, and transfers the resulting single finite gate sequence to SK. These are additional theorems, reconstructed in the long record, §9, "From the scalar formula to one ordinary QAOA word", from OpenAI (2026b), §§2–6 and Appendix A. Neither positive density nor (10) alone proves them. The selected construction gives no quantitative depth bound, finite-size rate or efficient angle-selection procedure.

6. Source and formal scope

This AI-assisted article reconstructs the cited results; it asserts no new theorem. The original analytic proofs are in OpenAI's manuscripts, with the prior mathematical sources identified above. The central imported input is the integrated estimate (5), including its analytic shape and approximation hypotheses.

Lean is a proof assistant: it checks proofs written in its formal language. For the support result, the linked files contain the proof implementation, theorem specification (challenge) and comparison-tool settings. These files concern measure and diffusion existence, full support, strict increase and both moment identities, conditional on a minimiser. The scalar implementation, challenge and configuration select a separate formal interface: its scalar and finite-Gaussian clauses assume a minimiser and an associated diffusion, while its ground-state/Parisi equality assumes only the minimiser. Their formal models have not been transported into one another in this record. Neither selected statement includes Corollary 3 or QAOA convergence.

The formal developments use Lean and its mathematical library, mathlib; we credit the system description by de Moura and Ullrich (2021) and The mathlib Community (2020). These scholarly references describe the tools; the accompanying source record identifies the exact versions used for the selected developments.

Source-integrity checks passed. After an earlier unavailable Comparator startup, later invocations built and exported the admitted full-support and scalar-value challenges. Both stopped during the implementing solution build because imported compiled modules were missing. No successful

replay of the corrected build configuration is recorded. Challenge export, tool-fixture success and the exact polynomial calculation establish no research-target acceptance, transitive axiom audit or independent human review. Exact versions, the failed attempts and remaining obligations are retained in the companion source review.

References

- de Moura, L. and Ullrich, S. (2021). "The Lean 4 Theorem Prover and Programming Language (System Description)." *CADE 28*. System description.
- The mathlib Community (2020). "The Lean Mathematical Library." *Certified Programs and Proofs (CPP '20)*. Library description.
- OpenAI (2026a). *Full support of the zero-temperature Sherrington–Kirkpatrick order parameter*. 27 September 2026. Manuscript, fixed revision. Theorem 1.1; §§2–6; Corollaries 6.4 and 7.1; Proposition 7.2.
- OpenAI (2026b). *QAOA attains the SK ground-state energy in the thermodynamic-first limit*. 25 September 2026. Manuscript, fixed revision. Theorem 1.1; §§2–6 and Appendix A.
- Auffinger, A. and Chen, W.-K. (2017). "Parisi formula for the ground state energy in the mixed p-spin model." *Annals of Probability* **45**, 4617–4631. Preprint v2: Theorem 1 and proof of Lemma 3.
- Auffinger, A., Chen, W.-K. and Zeng, Q. (2020). "The SK model is infinite step replica symmetry breaking at zero temperature." *Communications on Pure and Applied Mathematics* **73**, 921–943. Preprint v2: Theorem 1 and Remark 7.
- Chen, H.-B. (2026). *FRSB in the SK spin glass: convergence to full-interval support at zero temperature*. Version 1, 20 July 2026. Theorem 1.3; Propositions 4.2 and 4.6; Appendices B.6 and C.
- Lopatto, P. (2026). *Full replica symmetry breaking in the Sherrington–Kirkpatrick model*. arXiv version 3, revised 16 July 2026. Propositions 3.6, 7.1 and 9.1; related Cole–Hopf, density and crossing arguments.
- Sherrington, D. and Kirkpatrick, S. (1975). "Solvable Model of a Spin-Glass." *Physical Review Letters* **35**, 1792–1796. Original model.
- Jagannath, A. and Tobasco, I. (2016). "A dynamic programming approach to the Parisi functional." *Proceedings of the AMS* **144**, 3135–3150. Preprint v4. Positive-temperature control methods.
- Chen, W.-K., Handschy, M. and Lerman, G. (2018). "On the energy landscape of the mixed even p-spin model." *Probability Theory and Related Fields* **171**, 53–95. Preprint v4: Proposition 2, Theorem 5, Lemma 3 and Propositions 3–4.
- Auffinger, A. and Chen, W.-K. (2015). "On properties of Parisi measures." *Probability Theory and Related Fields* **161**, 817–850. Preprint v1: Lemma 2 and proof of Theorem 2(ii).
- Montanari, A. (2025). "Optimization of the Sherrington–Kirkpatrick Hamiltonian." *SIAM Journal on Computing* **54**, FOCS19-1–FOCS19-38. Preprint v2, 2019: §§2–3.
- El Alaoui, A., Montanari, A. and Sellke, M. (2021). "Optimization of mean-field spin glasses." *Annals of Probability* **49**, 2922–2960. Preprint v1: §§5–6.
- Farhi, E., Goldstone, J. and Gutmann, S. (2014). *A Quantum Approximate Optimization Algorithm*. Preprint v1. Algorithm definition; §VI concerns the fixed-instance depth limit.
- Farhi, E., Goldstone, J., Gutmann, S. and Zhou, L. (2022). "The Quantum Approximate Optimization Algorithm and the Sherrington–Kirkpatrick Model at Infinite Size." *Quantum* **6**, 759. Preprint v4.
- Basso, J., Farhi, E., Marwaha, K., Villalonga, B. and Zhou, L. (2022). "The Quantum

Approximate Optimization Algorithm at High Depth for MaxCut on Large-Girth Regular Graphs and the Sherrington–Kirkpatrick Model." *TQC 2022*, 7:1–7:21. Full preprint v3: transfer theorem and §7, equation (7.2); the full preprint has separate numbering.